

Problem

Obtain the Laplace transform of each of the following functions:

(a) $e^{-2t} \cos 3tu(t)$

(b) $e^{-2t} \sin 4 tu(t)$

(c) $e^{-3t} \cosh 2tu(t)$

(d) $e^{-4t} \sinh tu(t)$

(e) $te^{-t} \sin 2tu(t)$

Step-by-step solution

Step 1 of 5

(a)

Consider the signal $x(t) = e^{-2t} \cos 3tu(t)$.

Write the Laplace transform pairs that are used to find $X(s)$.

$$\cos at u(t) \xrightarrow{\mathcal{L}} \frac{s}{s^2 + a^2}$$

$$e^{-at} u(t) \xrightarrow{\mathcal{L}} \frac{1}{s + a}$$

Apply the Laplace transform pairs to find $X(s)$.

$$\cos 3tu(t) \xrightarrow{\mathcal{L}} \frac{s}{s^2 + 3^2}$$

$$e^{-2t} \cos 3tu(t) \xrightarrow{\mathcal{L}} \frac{s + 2}{(s + 2)^2 + 3^2}$$

Therefore, the Laplace transform $F(s)$ is $\boxed{\frac{s + 2}{(s + 2)^2 + 9}}$.

Step 2 of 5

(b)

Consider the signal $x(t) = e^{-2t} \sin 4tu(t)$.

Write the Laplace transform pairs that are used to find $X(s)$.

$$\sin at u(t) \xleftrightarrow{\mathcal{L}} \frac{a}{s^2 + a^2}$$

$$e^{-at} u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s + a}$$

Apply the Laplace transform pairs to find $X(s)$.

$$\sin 4tu(t) \xleftrightarrow{\mathcal{L}} \frac{4}{s^2 + 4^2}$$

$$e^{-2t} \sin 4tu(t) \xleftrightarrow{\mathcal{L}} \frac{4}{(s+2)^2 + 4^2}$$

Therefore, the Laplace transform $F(s)$ is $\boxed{\frac{4}{(s+2)^2 + 16}}$.

Step 3 of 5

(c)

Consider the signal $x(t) = e^{-3t} \cosh 2tu(t)$.

Write the Laplace transform pairs that are used to find $X(s)$.

$$\cosh tu(t) \xleftrightarrow{\mathcal{L}} \frac{s}{s^2 - a^2}$$

$$e^{-at} u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s + a}$$

Apply the Laplace transform pairs to find $X(s)$.

$$\cosh 2tu(t) \xleftrightarrow{\mathcal{L}} \frac{s}{s^2 - 2^2}$$

$$e^{-3t} \cosh 2tu(t) \xleftrightarrow{\mathcal{L}} \frac{s+3}{(s+3)^2 - 2^2}$$

Therefore, the Laplace transform $F(s)$ is $\boxed{\frac{s+3}{(s+3)^2 - 4}}$.

Step 4 of 5

(b)

Consider the signal $x(t) = e^{-4t} \sinh tu(t)$.

Write the Laplace transform pairs that are used to find $X(s)$.

$$\sinh tu(t) \xleftrightarrow{\mathcal{L}} \frac{a}{s^2 - a^2}$$

$$e^{-at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+a}$$

Apply the Laplace transform pairs to find $X(s)$.

$$\sinh tu(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s^2 - 1^2}$$

$$e^{-4t} \sin 4tu(t) \xleftrightarrow{\mathcal{L}} \frac{1}{(s+4)^2 - 1}$$

Therefore, the Laplace transform $F(s)$ is $\boxed{\frac{1}{(s+4)^2 - 1}}$.

Step 5 of 5

(e)

Consider the signal $x(t) = te^{-t} \sin 2tu(t)$.

Write the Laplace transform pairs that are used to find $X(s)$.

$$\sin atu(t) \xleftrightarrow{\mathcal{L}} \frac{a}{s^2 + a^2}$$

$$e^{-at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+a}$$

$$tf(t) \xleftrightarrow{\mathcal{L}} -\left(\frac{d}{ds}F(s)\right)$$

Apply the Laplace transform pairs to find $X(s)$.

$$\sin 2tu(t) \xleftrightarrow{\mathcal{L}} \frac{2}{s^2 + 2^2}$$

$$e^{-t} \sin 2tu(t) \xleftrightarrow{\mathcal{L}} \frac{2}{(s+1)^2 + 2^2}$$

$$te^{-t} \sin 2tu(t) \xleftrightarrow{\mathcal{L}} -\left(\frac{d}{ds}\left(\frac{2}{(s+1)^2 + 2^2}\right)\right)$$

$$= -\left(\frac{-2 \times 2(s+1)}{\left((s+1)^2 + 4\right)^2}\right)$$

$$= \frac{4(s+1)}{\left[(s+1)^2 + 4\right]^2}$$

Therefore, the Laplace transform $F(s)$ is, $\boxed{\frac{4(s+1)}{\left[(s+1)^2 + 4\right]^2}}$.